

In [ ]:

```
import pandas as pd
import mysql.connector
import os

# List of CSV files and their corresponding table names
csv_files = [
    ('customers.csv', 'customers'),
    ('orders.csv', 'orders'),
    ('sellers.csv', 'sellers'),
    ('products.csv', 'products'),
    ('geolocation.csv', 'geolocation'),
    ('payments.csv', 'payments') , # Added payments.csv for specific handling
    ('order_items.csv', 'order_items')
]

# Connect to the MySQL database
conn = mysql.connector.connect(
    host='localhost',
    user='root',
    password='PASSWORD@629',
    database='ecommerce'
)
cursor = conn.cursor()

# Folder containing the CSV files
folder_path = 'C:/Users/User/Desktop/Data Analytics/All projects/Python + SQL project/Da

def get_sql_type(dtype):
    if pd.api.types.is_integer_dtype(dtype):
        return 'INT'
    elif pd.api.types.is_float_dtype(dtype):
        return 'FLOAT'
    elif pd.api.types.is_bool_dtype(dtype):
        return 'BOOLEAN'
    elif pd.api.types.is_datetime64_any_dtype(dtype):
        return 'DATETIME'
    else:
        return 'TEXT'

for csv_file, table_name in csv_files:
    file_path = os.path.join(folder_path, csv_file)

    # Read the CSV file into a pandas DataFrame
    df = pd.read_csv(file_path)

    # Replace NaN with None to handle SQL NULL
    df = df.where(pd.notnull(df), None)

    # Debugging: Check for NaN values
    print(f"Processing {csv_file}")
    print(f"NaN values before replacement:\n{df.isnull().sum()}\n")

    # Clean column names
    df.columns = [col.replace(' ', '_').replace('-', '_').replace('.', '_') for col in df.columns]

    # Generate the CREATE TABLE statement with appropriate data types
    columns = ', '.join([f'`{col}` {get_sql_type(df[col].dtype)}' for col in df.columns])
```

```

create_table_query = f'CREATE TABLE IF NOT EXISTS `{table_name}` ({columns})'
cursor.execute(create_table_query)

# Insert DataFrame data into the MySQL table
for _, row in df.iterrows():
    # Convert row to tuple and handle NaN/None explicitly
    values = tuple(None if pd.isna(x) else x for x in row)
    sql = f"INSERT INTO `{table_name}` ({', '.join(['`'+col+'`' for col in df.co
    cursor.execute(sql, values)

# Commit the transaction for the current CSV file
conn.commit()

# Close the connection
conn.close()

```

Processing customers.csv  
 NaN values before replacement:

|                          |       |
|--------------------------|-------|
| customer_id              | 0     |
| customer_unique_id       | 0     |
| customer_zip_code_prefix | 0     |
| customer_city            | 0     |
| customer_state           | 0     |
| dtype:                   | int64 |

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